Equal Temperament Pitch & Frequency Conversions

Mathematical Models of [mtof] and [ftom] Nodes

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Abstract

This document formulates the logarithmic and exponential relationships between MIDI Note numbers $m \in [0, 127]$, fundamental frequencies $f \in \mathbb{R}^+$ (in Hz), and microtonal cent tuning ratios.

1 MIDI Note to Frequency Conversion ([mtof])

In 12-tone equal temperament (12-TET) with concert pitch reference $A_4 = 440\,\text{Hz}$ at MIDI Note $m = 69$, the frequency $f(m)$ is defined exponentially:

$$f(m) = 440 \cdot 2^{\frac{m-69}{12}} \tag{1}$$

For an arbitrary reference frequency $A_4 = f_0$:

$$f(m, f_0) = f_0 \cdot 2^{\frac{m-69}{12}} \tag{2}$$

2 Frequency to MIDI Note Conversion ([ftom])

The inverse mapping from frequency f to MIDI Note number $m(f)$ is derived by taking the base-2 logarithm:

$$m(f) = 69 + 12 \cdot \log_2\left(\frac{f}{440}\right) \tag{3}$$

Using natural logarithms:

$$m(f) = 69 + \frac{12}{\ln 2} \cdot \ln\left(\frac{f}{440}\right) \tag{4}$$

3 Cent Tuning & Transposition Ratios

The frequency ratio $R(c)$ for a microtonal interval of c cents ($100\,\text{cents} = 1\,\text{semitone}$) is:

$$R(c) = 2^{\frac{c}{1200}} \tag{5}$$